Lecture 2: Concept of Global Reference Systems and Reference Frames

Hierarchy of Systems/Frames

Celestial

1. BCRS (Barycentric Celestial Reference System)
    - ICRS/ICRF (VLBI, GAIA), HCRF (Hipparcos)
2. GCRS (Geocentric Celestial RS/RF)

Terrestrial

1. Global (ITRS/ITRF, WGS84)
2. Regional frame
3. National frame

Time

UTC, TAI

Definition: Reference System and Frame

- RS: A **theoretical** concept and definition of system
- RF: Physical and numerical **realisation** with markers and coordinates through **time**

Definition: Coordinate System

Specifies a **mathematical method** for locating points within a particular reference frame

1. Ellipsoidal (h, φ, λ)
2. Spherical (r, θ, λ)
3. Cartesian (x, y, z)
4. Topocentric (δ, α), or (altitude, azimuth)

Terrestrial Reference System (TRS)

- Spatial reference system
- co-rotating with the Earth (in diurnal motion in space)

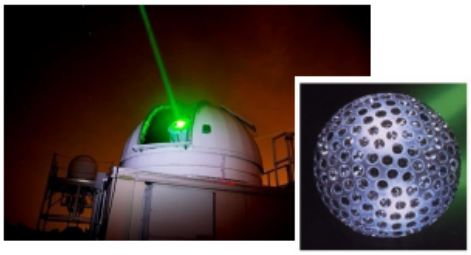
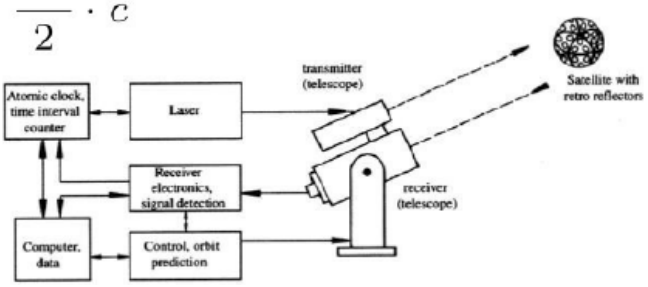
- positions attached to solid surface of Earth:
  - Coordinate variations with time (geophysical effects: tectonic/tidal deformation etc.)
- Conventional (C)TRS: Defined by
  - Set of all conventions, algorithms, and constants
    - Provide: Origin, Scale and Orientation
    - For: system and time evolution (IUGG Resolution)
  - Origin: Geocentre w.r.t. total mass of Earth system (solid Earth + oceanic and atmospheric masses)
  - Unit: Meter
  - References:
    - Reference Velocity Field (crustal dynamic)
    - Reference epoch
    - Reference ellipsoid with GRS80 dimension

Terrestrial Reference Frame (TRF)

- Set of physical points with precisely determined coordinates in a specific coordinate system attached to a TRS
  - Realisation of TRS
  -
- Defined by astronomical and geodetic communities (IAU, IUGG, IERS, ...)
- (C)TRF: set of physical points with precisely determined coordinates in a specific coordinate system
  - as realisation of ideal TRS
  - 2 Types: **dynamical** CTRF and **kinematical** CTRF
    - If dynamical model is applied in the process of determining these coordinates

Individual TRFs

- TRF solutions computed by IERS analysis centers
- Observations of space geodesy techniques **(...-based TRF)**
  - **VLBI** - Very Long Baseline Interferometry
    - since 80s
    - IVS network: 34sites
    - usage of astronomical radio source (Quasar)
      - phase offset -> distance
    - Results:

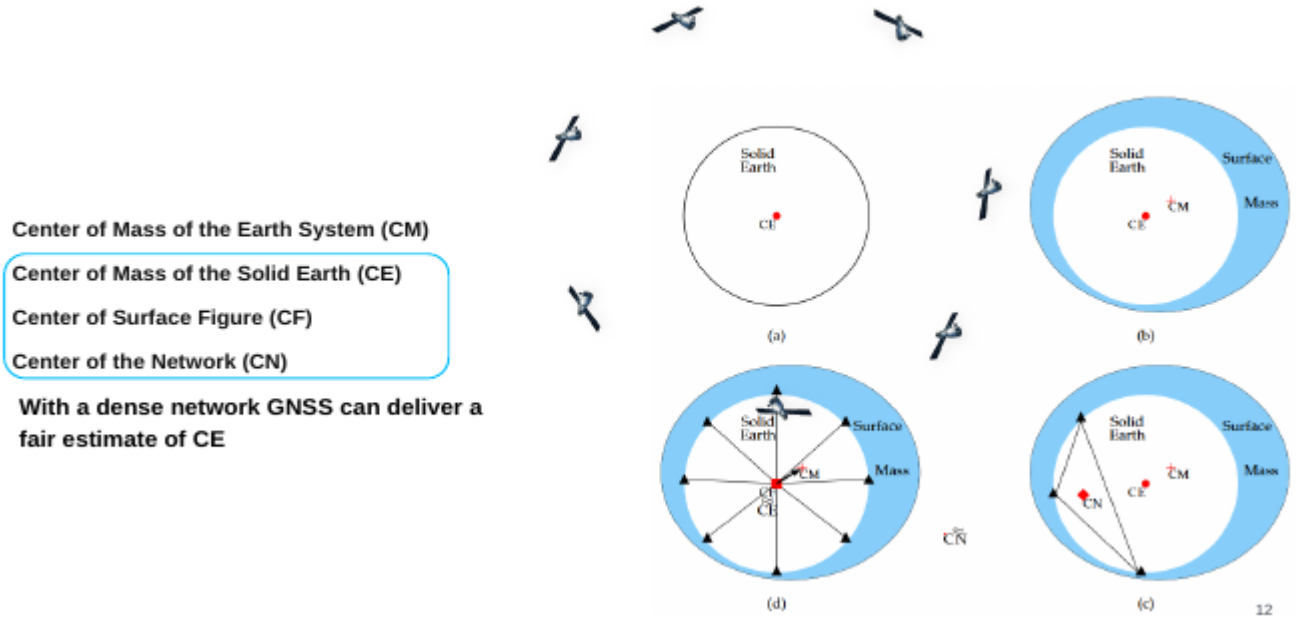
    1. Earth Orientation Parameters
    2. Absolute 3D coordinates after tied to CRF
    3. Scale (Size) and orientation of the reference frame
- **SLR** - Satellite Laser Ranging
  - since 74
  - ILRS network: 39 sites
  - 
    $$s = \frac{\Delta t}{2} \cdot c$$
    

  - RTT from laser
  - Results:
    1. Earth Orientation Parameters
    2. Absolute 3D coordinates
    3. Earth's center of mass (geocentre)
    4. Scale of TRF (SLR-based TRF)
  - Defines **Origin** (geoocentre) and with VLBI the **scale**
- **DORIS** - Doppler Orbitography and Radiopositioning Integrated by Satellite
  - since 60 - French
  - IDS network: 60 sites
  - Doppler shift of signal sent from Ground-based beacons
  - Results:
    1. Earth Orientation Parameters
    2. Absolute 3D coordinates
       - PoD (TOPEX/Poseidon, Jason 1/2/3)
  - Benefit:
    1. **Global coverage** including oceans and *remote* regions
    2. Dual use for precise **satellite orbit determination**
    3. Continuous, automatic data collection
    4. Compact and robust ground equipment
- **GNSS**
  - since 80
  - IGS network: 500 sites

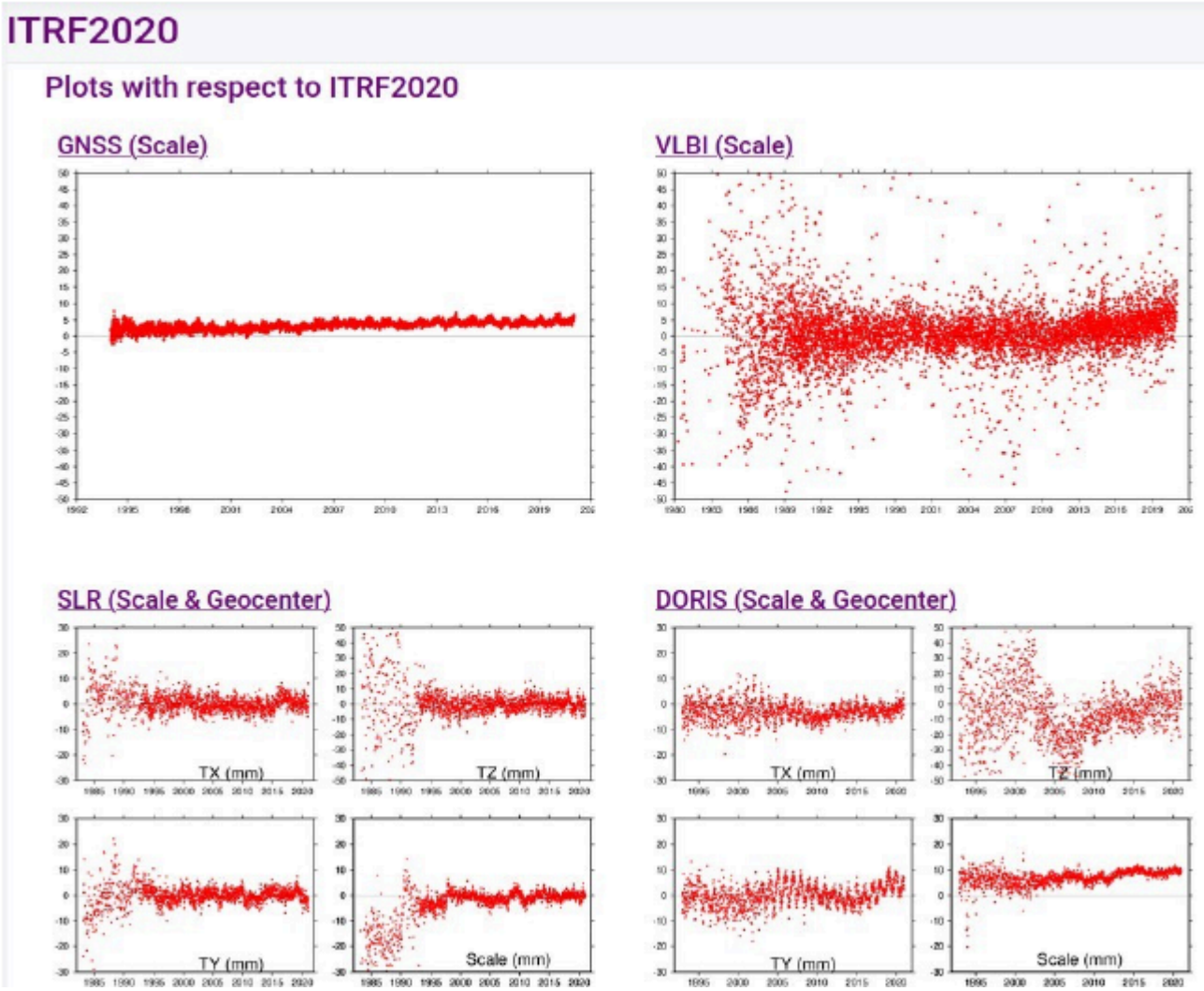
- Results:
  1. Earth Orientation Parameters (Polar motion and LoD)
  2. Absolute 3D Coordinates
- Benefit:
  1. Dense global network
  2. Continuous temporal coverage:
  3. High precision in coordinates
  4. Budget friendly

Geocenter motion

GNSS: subject to complex nongravitational dynamics
-> limits dynamic interpretation of station motions in the CM-frame in terms of gloabl mass redistribution.



Scale and Origin Time Series (ITR2020)



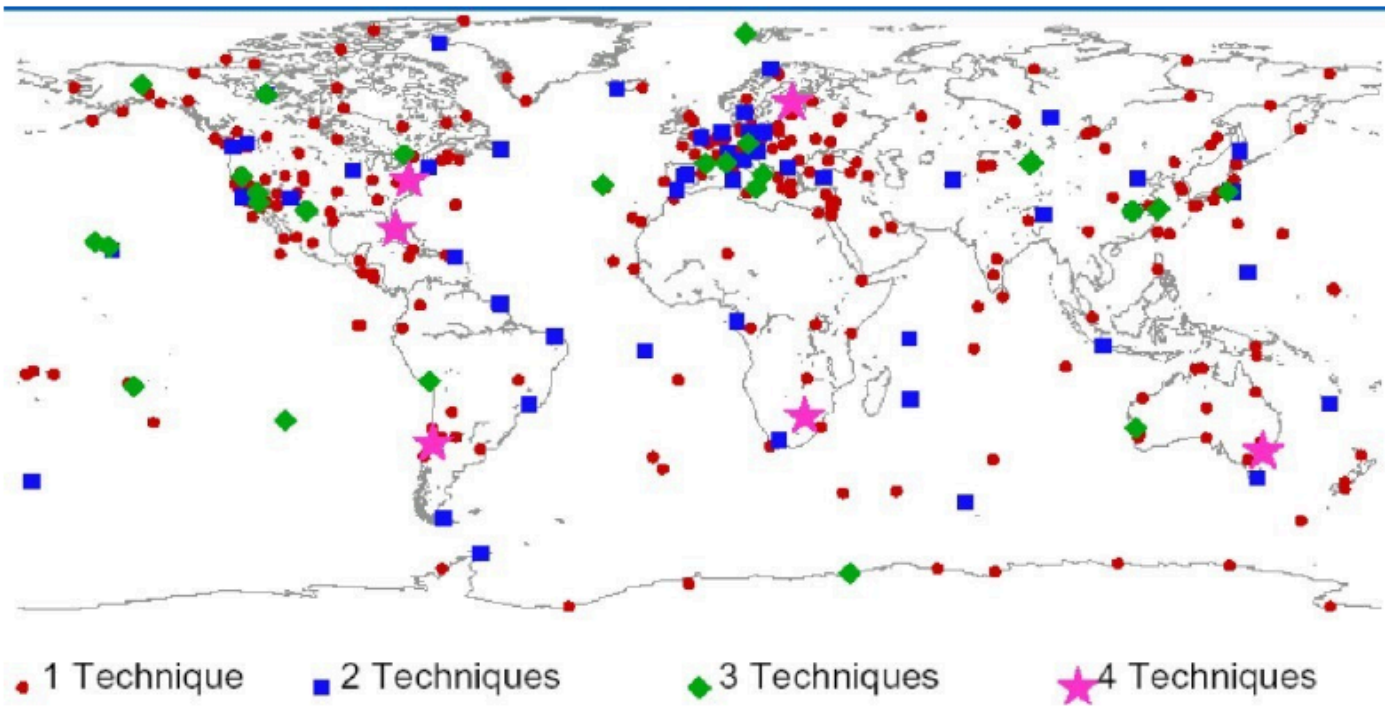
Delivery of each Technique

EOPs | Datum parameters

Technique	Station positions	Pole positions	UT1	LOD	Nutation parameters	origin	scale
GNSS	√	√	×	×	×	×	×
DORIS	√	√	×	×	×	×	×
SLR	√	√	×	×	×	√	√
VLBI	√	√	√	√	√	×	√

-> Combination of **all** space geodetic **techniques** for definition of **all geodetic parameters** (accurate TRF)

Co-located sites

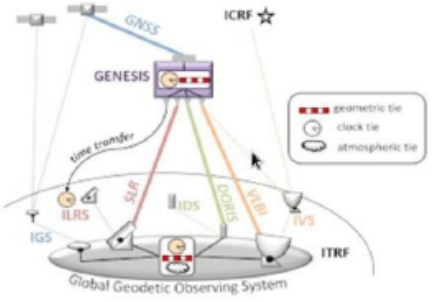


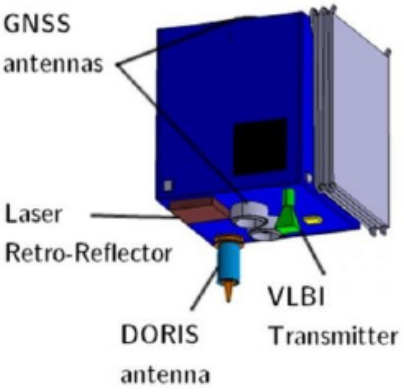
For connection of TRFs: some ties between frames

- Local Ties in co-location sites
  - Local surveys (total station): link techniques at co-location sites using either classical geodesy or GPS
- Spacial Ties - Multi-technique spacecraft
  - e.g. GNSS satellites equipped with SLR reflectors

GENISIS co-location of all geodetic techniques in space

- ESA initiative - launch: 2027
  - TRF with accuracy of 1mm & long-term stability of 0.1 mm/y
  - VLBI transmitter
  -



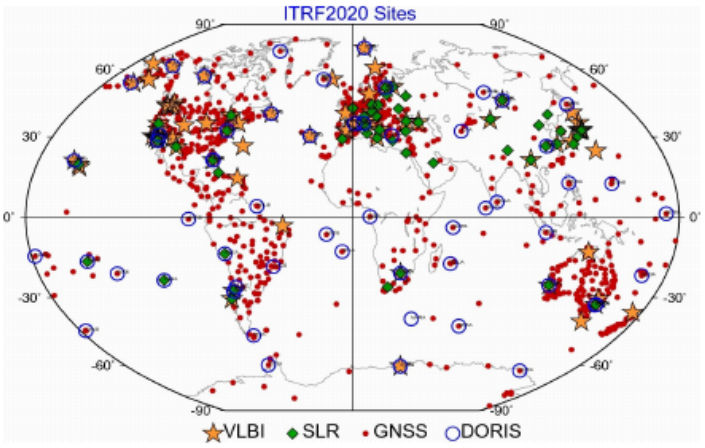


ITRF

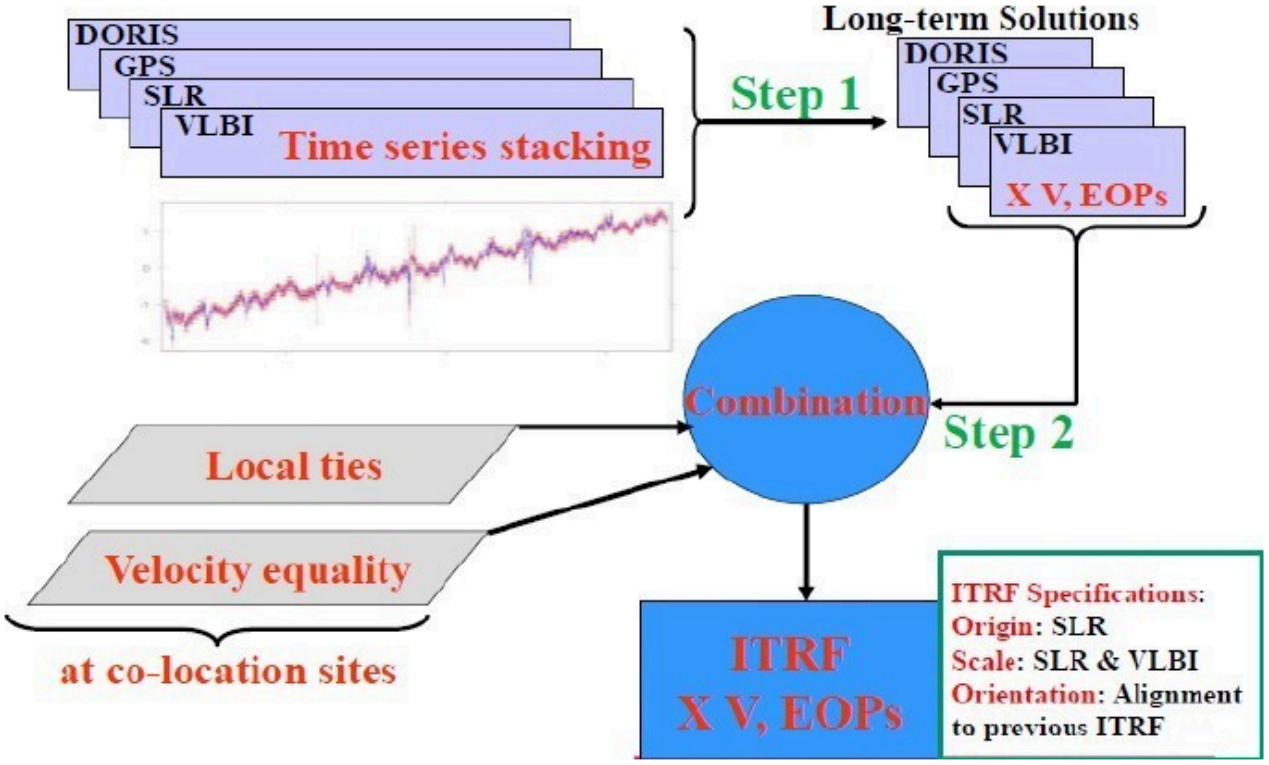
- **Combination** of individual TRF solutions
  - Computed by IERS analysis centers using all observations

- Individual TRF solution:
  - Position and velocity
  - Full variance matrices
  - In **SINEX** format (Solution INdependent Exchange)

Locations of stations for each Individual techniques that are used to produce ITRF.



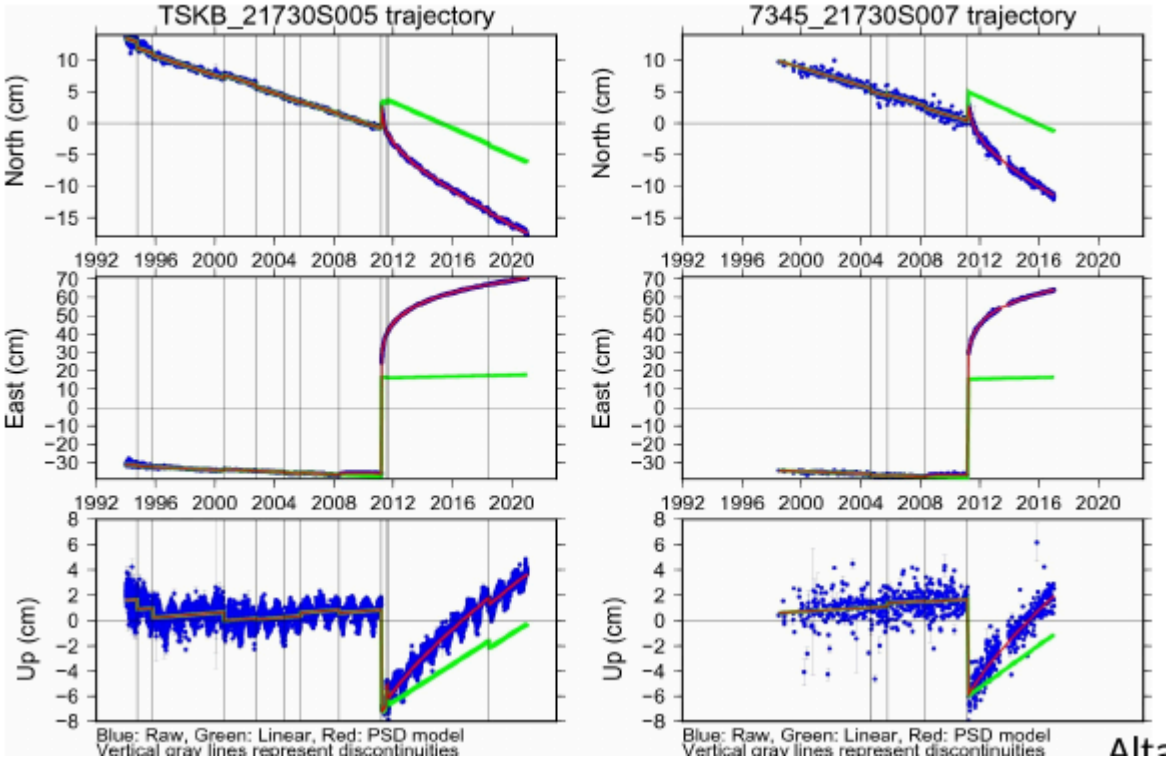
- https://itrf.ign.fr/en/network/map



- ni et al. (2023) Combination of solutions

Name	data until Ende	reference date	
BTS84	1984	01.01.1984 (without station velocities)	27 SLR, 13 VLBI (Precursor of ITRF)
...	...	...	
ITRF88	1988	01.01.1988	SLR, VLBI, w/o GPS points (96)
ITRF89	1989	01.01.1988	SLR, VLBI, w/o GPS points (117)
...	...	...	
ITRF91	1991	01.01.1988	SLR, VLBI, 14 GPS points **with estimated velocities**
ITRF92	1992	01.01.1988	SLR, VLBI, 40 GPS points
...	...	...	
ITRF97	1997	01.01.1997	SLR, VLBI, > 140 GPS points
ITRF2000	2000	01.01.1997	SLR, VLBI, DORIS, > 300 GPS points
ITRF2005	2005	01.01.2000	53 SLR, 87 VLBI, 64 DORIS, > 450 GPS points
ITRF2008	2008	01.01.2005	87 SLR, 103 VLBI, 66 DORIS, 560 GPS
ITRF2014	2014	01.01.2010	975 stations with 1499 observing platforms (1054 GPS)

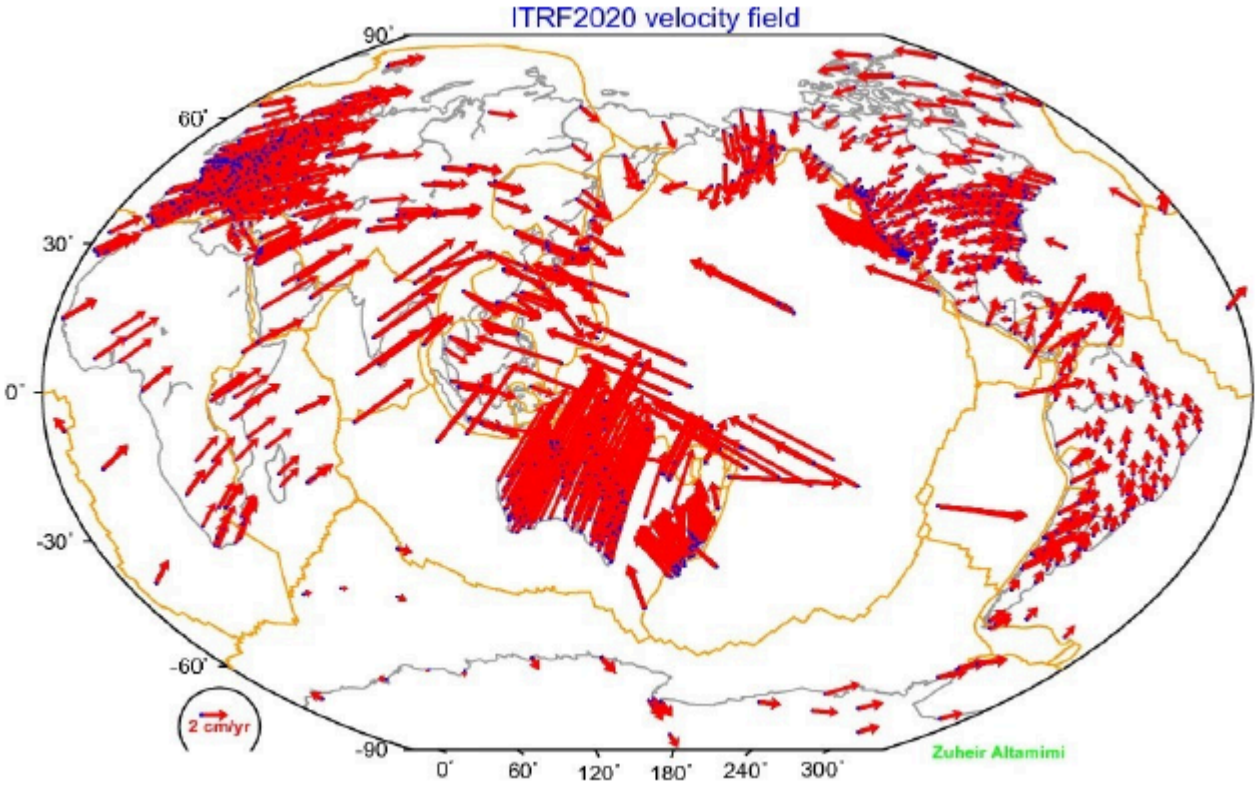
Time evolution of ITRF

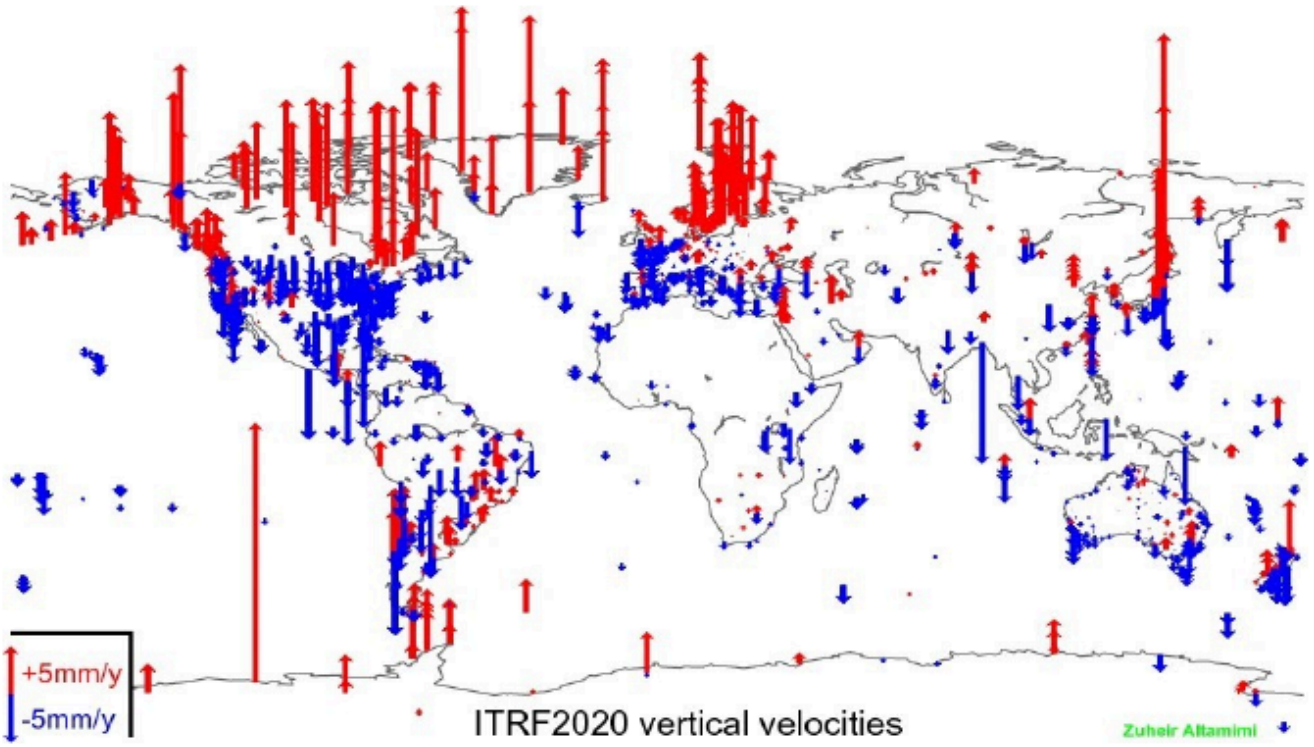
- Due to
  - Earthquakes
  - Non-linear motion
- 

- Update related to measurement techniques
  - Model updates for GNSS, DORIS, etc
  - SLR range biases
  - VLBI antenna deformation
  - Improved data processing e.g. loading, geocenter motion, etc

- More stations and longer time series since last update

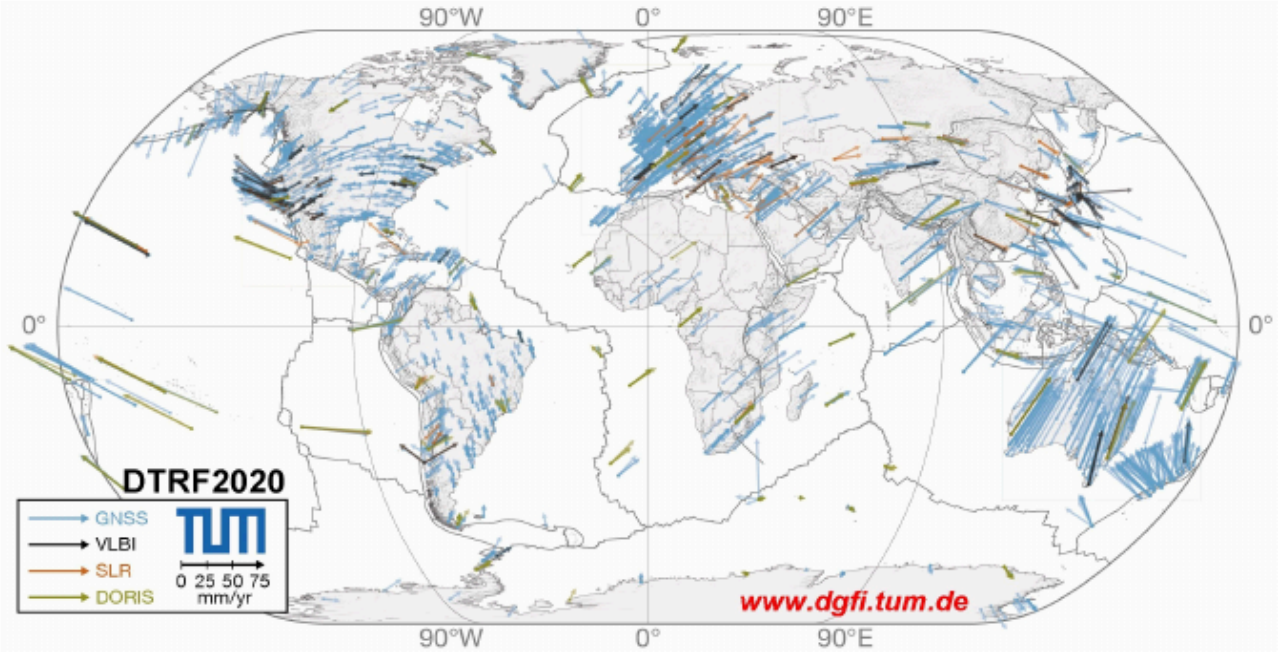
ITRF velocities

- ITRF is crust-based TRF and is mainly used for positions of points fixed on solid earth crust
- 

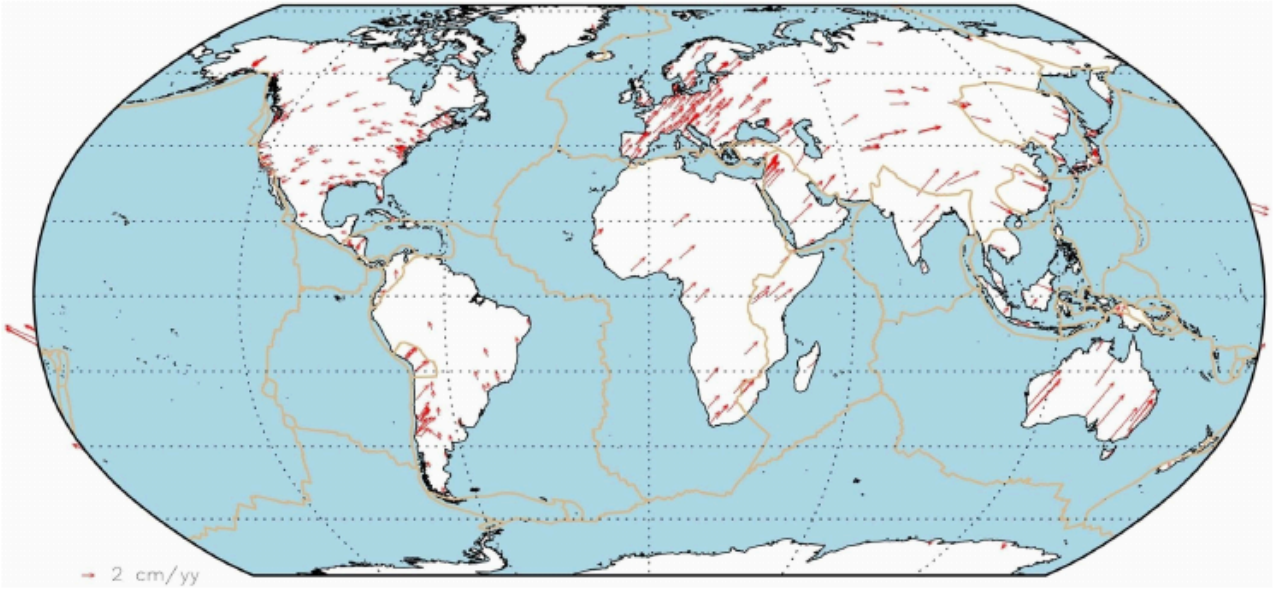
- 

DTRF

- DGFI-TUM responsible for DTRF (latest update 2020)
  - Based on VLBI, SLR; DORIS and GPS

- Crust-based TRF
- Different combination strategies
- 

JTRF

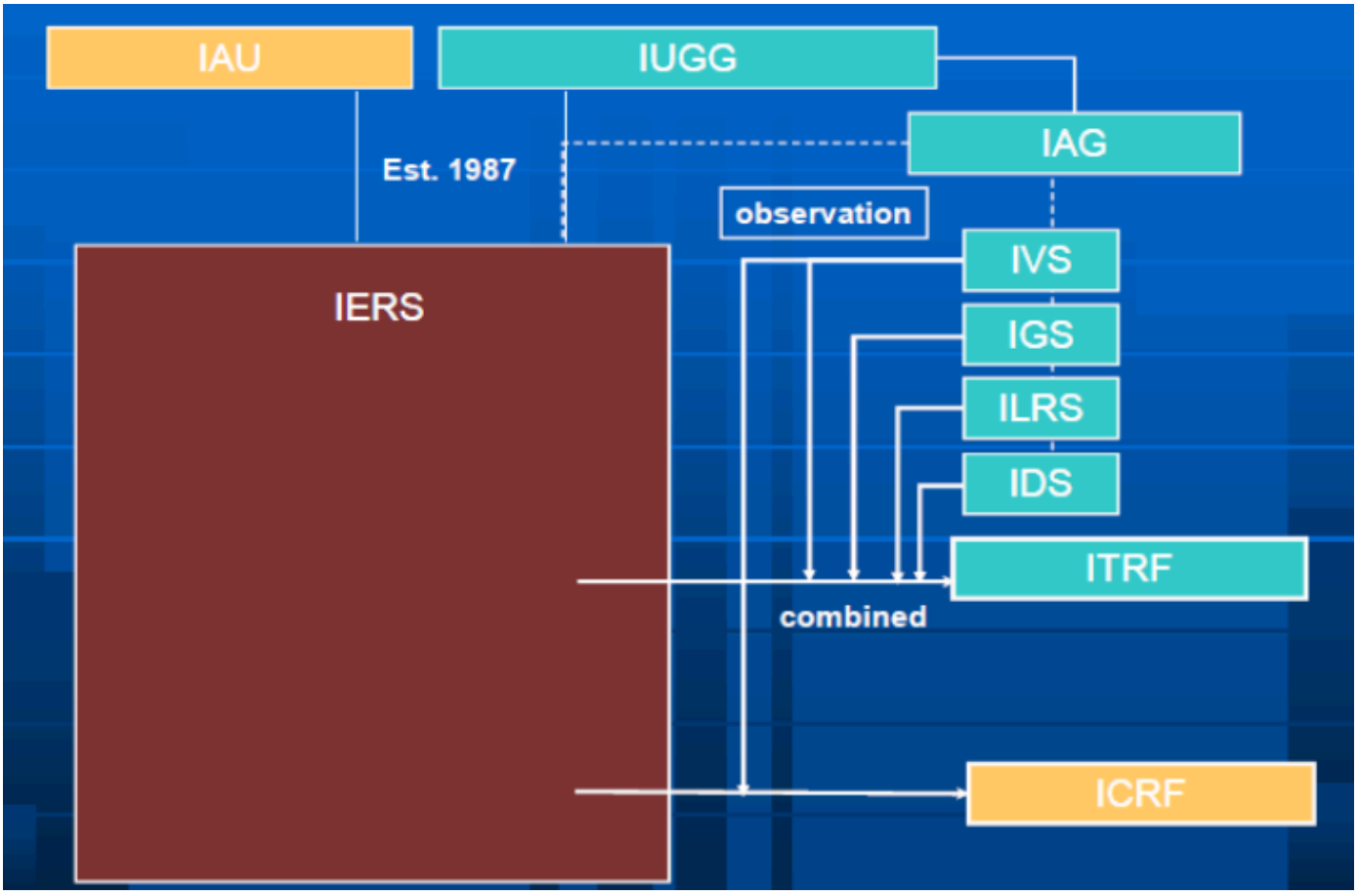
- NASA-JPL responsible for JTRF (latest update 2020)
  - Based on VLBI, SLR, DORIS and GPS
  - Crust-based TRF
  - 

ITRF2014 STATION POSITIONS AT EPOCH 2010.0 AND VELOCITIES

DOMES NB.	SITE NAME	TECH. ID.	X|Vx	Y|Vy	Z|Vz	Sigmas			SOLN	DATA_START	DATA_END
			m|m/y								
14201M004	WETTZELL	SLR 7596	4075582.288	931837.412	4801560.173	0.023	0.043	0.026			
14201M004			-.0154	0.0172	0.0112	.0001	.0001	.0001			
14201S100	WETTZELL	VLBI 7593	4075572.557	931755.259	4801584.444	0.001	0.001	0.001			
14201S100			-.0160	0.0171	0.0101	.0001	.0000	.0001			
14209S001	EFFELSBERG	VLBI 7203	4033947.353	486990.646	4900430.889	0.003	0.001	0.003			
14209S001			-.0138	0.0169	0.0110	.0002	.0001	.0003			
14209S001	EFFELSBERG	VLBI 7203	4033947.366	486990.652	4900430.902	0.001	0.001	0.002	2	96:275:00	00:000:00
14209S001			-.0138	0.0169	0.0110	.0002	.0001	.0003			
14201M010	WETTZELL	GNSS WTZR	4075580.553	931853.797	4801568.136	0.001	0.001	0.001			
14201M010			-.0160	0.0171	0.0101	.0001	.0000	.0001			
10003M004	Toulouse	GNSS TOUL	4627846.029	119629.333	4372999.818	0.001	0.001	0.001			
10003M004			-.0114	0.0193	0.0121	.0001	.0000	.0001			

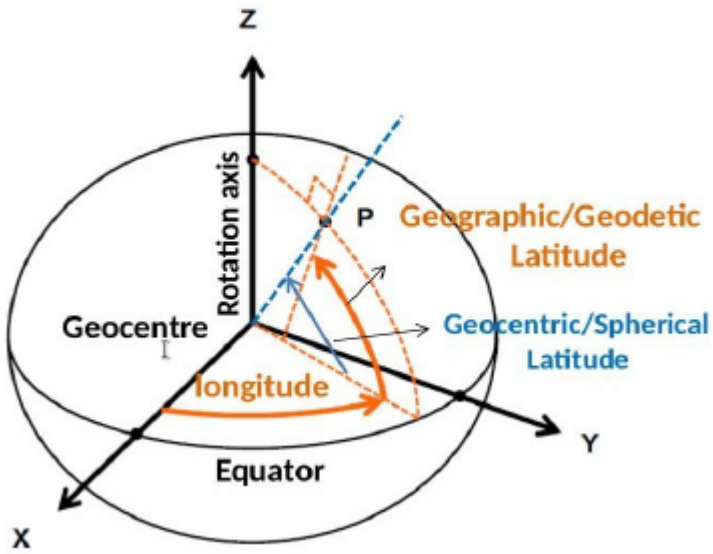
$$X_t = X_{t_0} + v_x \cdot (t - t_0)$$

Organisations

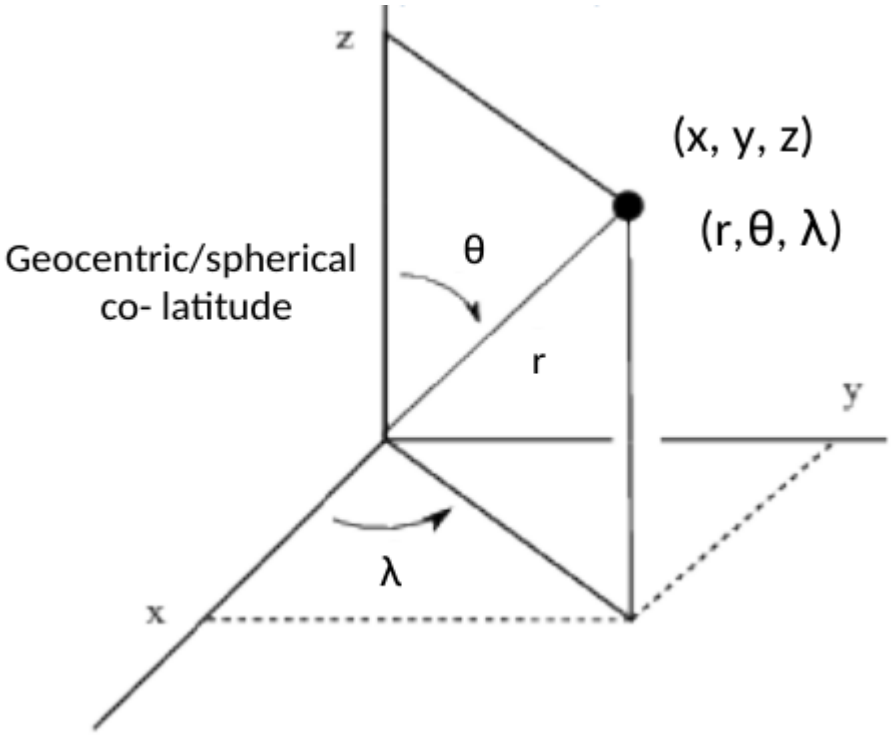


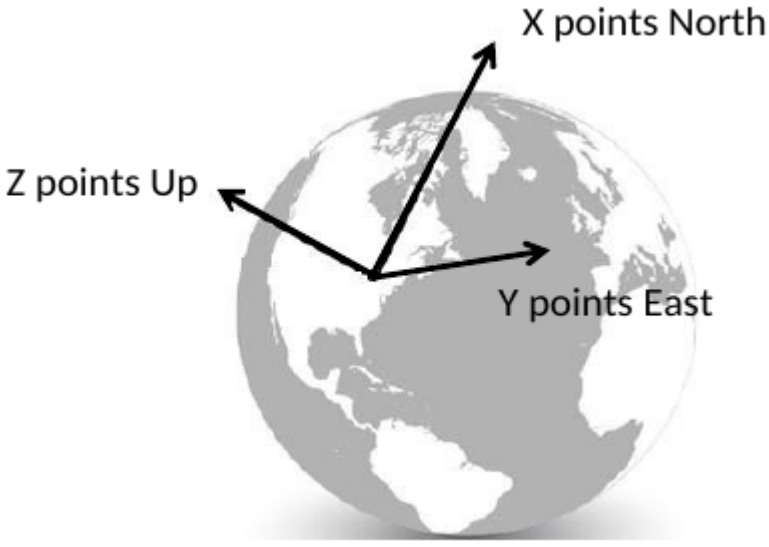
- IERS - International Earth Rotation and Reference System
  - Global reference to the astronomical, geodetic and geophysical communities
- TRF solutions computed by IERS analysis centers using space geodesy observations

Coordinate Systems

- **Ellipsoidal**:
  - 

  - Tied to reference ellipsoid
  - Reference meridian (x-axis) at Greenwich Convention 1880
  - Examples: TRS
    - TRFs, GRS80, ... ,
    - WGS84:
      - Datum used by GPS system
      - Defined by coordinates of tracking and control stations
      - Regularly aligned with ITRF
      - Latest realisation WGS84 is G2139 aligned to ITRF14
      - Similar to ITRF:
        - Provide crust-based results
      - But
        - Maintained by different organisations
        - different update cycles
        - Different purposes
- **Spherical and Cartesian**:

- 

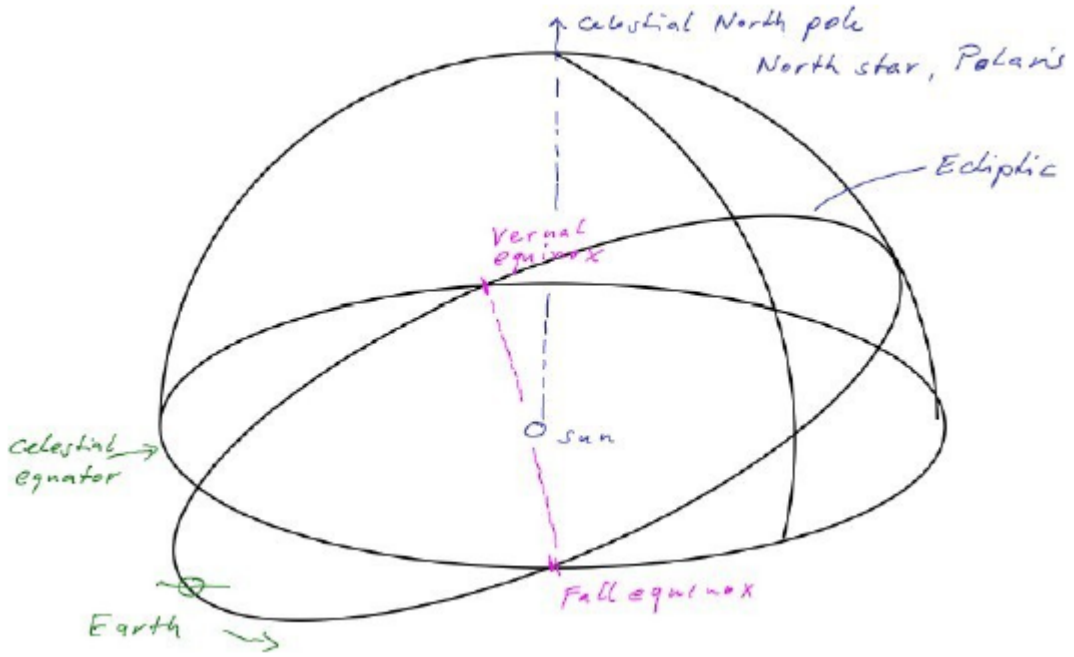
  - Examples:
    - TRS and CRS: TRFs and CRFs (Dec., R.A.).
- **Topocentric**
  - 

  - No standard definition
  - Z-Axis is normal to reference surface
    - Parallel to gravity vector
  - XY-plane: observers's local horizon
  - Left-handed coordinate system
  - Coordinates: Altitude (elevation angle), Azimuth
  - Examples:
    - Conventional surveying
    - Astro-geodesy

- Satellite Geoesy ...

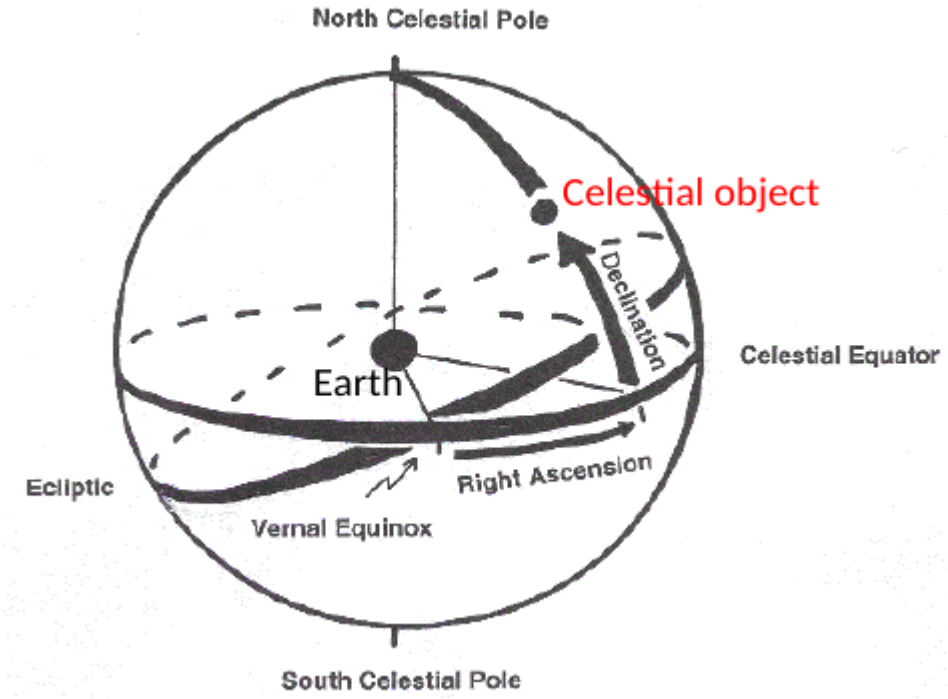
CRS - Celestrial Reference System

CRS: theoretical concept for spatial reference system of coordinates axes w.r.t positions of celestial bodies



Coordinates in CRS

1. Declination (Dec.: $\delta \in [-90^\circ\ 90^\circ)$)
2. Right Ascension (R.A.: $\alpha \in [0h, 24h)$)



CRF

- Set of physical objects with precisely determined coordinates in specific coordinate system attached to CRS
- 2 types: Dynamical and Kinematical CRF
  - if dynamic model is used for coordinate determination
  - 1991 IAU CRF based on kinematical defintion
    - International CRS (ICRS)
- Usage of Quasars, not stars
- Coordinate axes fixed w.r.t Universe (no global rotation)

Techniques to realise CRF: VLBI

- ICRF realises ICRS with precise coordinates of extra galactic radio sources observed by VLBI
- (Radio Celestial Frames)

Techniques to realise CRF: GAIA satellite

- contains 2 optical telescopes:
  - precise location determination of stars, distance, velocities
- Slow spinning: spinning telescopes across entire celestial sphere
- Can also be used to **realise CRF**:
  - GAIA CRF2 (556,869 quasars - compared to 4536 in ICRF3 based on VLBI)
  - realised optical based ICRF
  - replaced older Hipparcos Celestrial RF HCRF

VLBI is a crucial technique

- VLBI observations tie *antenna positions* in **ITRF** to distance *celestial sources* in **ICRF**
- Linking TRFx to CRFs through **EOPs** (nutation and Earth rotation parameters and Earth's angular velocity (UT1-UTC)).